Design Verification Plan & Report (virtual-test version) — V11 (t1_r3)

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Note: virtual verification only; values mechanically taken from simulation outputs. Items beyond the pure-simulation boundary are honestly marked N/A.

#	Verification item	Condition	Result value	Determination	Source
1	1C discharge capacity	1C CC to 2.5 V, 25 °C (SPMe)	5.0849 Ah	informational (no capacity threshold registered in entry 0)	`r5_v11_1c_spme.json:capacity_ah`
2	Energy density	1C discharge energy ÷ contract mass	605.33 Wh/kg	✓ PASS (≥ 392.61)	`r5_v11_energy.json:energy_density_wh_kg`
3	4C fast-charge temperature rise	4C CC charge, 45 °C ambient, lumped thermal	T_max 321.2 K (48.0 °C)	✓ PASS (≤ 333.15 K)	`r5_v11_4c.json:T_max_K`
4	4C fast-charge plating	same run, anode surface potential	min +0.0425 V → plated = false	✓ PASS (no plating)	`r5_v11_4c.json:anode_potential_v`
5	4C charge acceptance	4C (20.34 A) to 4.2 V event	4.677 Ah = 92.0% of 1C capacity in 13.8 min	✓ task "support 4C fast charge" demonstrated	`r5_v11_4c.json:capacity_ah` × nominal 5.085
6	Voltage window	parameter set	2.5 – 4.2 V	✓ PASS	Chen2020 cut-offs
7	Overcharge abuse to 4.7 V	0.5C CC to 4.7 V (DFN) → coupled thermal-runaway ODE (mass 0.03076 kg)	v_end 4.700 V; T_max 298.57 K; triggered = false	✓ PASS (no thermal runaway)	`r5_v11_oc.json`, `r5_v11_tr.json`
8	Nail penetration	—	—	N/A (beyond pure simulation boundary, requires physical experiment)	—
9	Crush / drop	—	—	N/A (requires physical experiment)	—
10	Cycle life	—	—	N/A (no aging model run in this case)	—
11	Rate-pulse internal resistance (HPPC)	—	—	N/A (protocol not run)	—

Conclusion

All six registered verification items pass; item 1 is informational (no threshold registered). Uncovered items (8–11) are outside the pure-simulation boundary and are listed as paper limitations.